WhatNext Vision Motors

Salesforce CRM - Vehicle Order Management System

FINAL PROJECT REPORT

Project Demonstration Phase

Abstract

WhatNext Vision Motors is a Salesforce CRM solution that manages the complete vehicle ordering process of a dealership network. The system captures vehicle, dealer, customer, order, test-drive and service-request data in a structured data model, exposes it through a single Lightning application, and automates routine work using Record-Triggered Flows, Apex triggers, Batch Apex and Scheduled Apex. Dealers are assigned to new orders automatically, customers receive test-drive reminders, stock is validated before an order is accepted, and pending orders are confirmed in batches when stock is available.

1. Introduction

Vehicle dealerships handle a large number of customer enquiries, orders, test drives and service requests. When these are tracked manually, orders can be placed for vehicles that are out of stock, dealers are assigned late, and customers miss test-drive appointments. This project uses the Salesforce platform to build a CRM that connects these activities and reduces manual effort.

1.1 Objectives

- Create and activate the Salesforce Developer environment.
- Build the core data model for vehicles, dealers, customers, orders, test drives and service requests.
- Provide custom tabs and a centralized Lightning application.
- Define the fields and lookup relationships required for data capture and automation.
- Automate dealer assignment and test-drive reminders with Record-Triggered Flows.
- Use Apex to validate stock and update stock for confirmed orders.
- Process pending orders with Batch Apex and automate execution with Scheduled Apex.

2. Technology Used

Area	Salesforce Feature
Environment	Salesforce Developer Account
Data model	Custom Objects, Fields, Lookup Relationships
User interface	Custom Tabs, Lightning App Manager
Declarative automation	Record-Triggered Flows, Scheduled Path, Send Email action
Programmatic logic	Apex Trigger and Handler class, Batch Apex, Scheduled Apex

3. System Design

3.1 Development Flow

Salesforce Account → Objects → Tabs → Lightning App → Fields & Relationships → Auto Assign Dealer Flow → Test Drive Reminder Flow → Apex Trigger Handler → Batch Apex → Scheduled Apex

3.2 Data Model

Object	Purpose	Relationships
Vehicle__c	Stores vehicle details	Dealer and Orders
Vehicle_Dealer__c	Stores authorized dealer information	Orders
Vehicle_Customer__c	Stores customer details	Orders and Test Drives
Vehicle_Order__c	Tracks vehicle purchases	Customer and Vehicle
Vehicle_Test_Drive__c	Tracks test-drive bookings	Customer and Vehicle
Vehicle_Service_Request__c	Tracks vehicle servicing requests	Customer and Vehicle

3.3 Fields and Relationships

Object	Key Fields
Vehicle__c	Vehicle_Name__c (Text); Vehicle_Model__c (Picklist); Stock_Quantity__c (Number); Price__c (Currency); Dealer__c (Lookup); Status__c (Picklist)
Vehicle_Dealer__c	Dealer_Name__c (Text); Dealer_Location__c (Text); Dealer_Code__c (Auto Number); Phone__c (Phone); Email__c (Email)
Vehicle_Customer__c	Customer_Name__c (Text); Email__c (Email); Phone__c (Phone); Address__c (Text); Preferred_Vehicle_Type__c (Picklist)
Vehicle_Order__c	Customer__c (Lookup); Vehicle__c (Lookup); Order_Date__c (Date); Status__c (Picklist)
Vehicle_Test_Drive__c	Customer__c (Lookup); Vehicle__c (Lookup); Test_Drive_Date__c (Date); Status__c (Picklist)
Vehicle_Service_Request__c	Customer__c (Lookup); Vehicle__c (Lookup); Service_Date__c (Date); Issue_Description__c (Text); Status__c (Picklist)

A Lookup Relationship field is created on the Vehicle object that references the Vehicle Dealer object, so every vehicle record can point to its associated dealer. Orders, test drives and service requests reference both the customer and the vehicle.

4. Implementation

4.1 Milestone 1 - Salesforce Account

A Salesforce Developer account was created from the Developer signup page (name, email, role as Developer, college name, country India, postal code and username). The account was activated through the verification email by setting a password and security answer, which gave access to Salesforce Setup.

4.2 Milestone 2 - Objects

Custom objects were created from Setup → Object Manager → Create → Custom Object, with Allow Reports and Allow Search enabled. For example, the Vehicle object uses the label Vehicle, plural label Vehicles and a text record name. The Vehicle Dealer, Customer, Order, Test Drive and Service

Request objects were created in the same way.

4.3 Milestone 3 - Tabs

Custom Object Tabs were created for every project object from Setup → Tabs, so users can reach the records directly from navigation.

4.4 Milestone 4 - Lightning App

A Lightning app named **WhatNext Vision Motors** was created in App Manager. Its navigation contains Vehicle, Dealer, Customer, Order, Test Drive, Service Request, Reports and Dashboard, and the System Administrator profile was added.

4.5 Milestone 5 - Fields and Relationships

The fields listed in section 3.3 were created for each object, including picklist values, and the lookup from Vehicle to Dealer was configured.

4.6 Milestone 6A - Auto Assign Dealer Flow

A Record-Triggered Flow on Vehicle Order runs when a record is created and Status__c equals Pending. It gets the customer record using the customer ID, finds the nearest dealer from the customer address or location, and updates the order with that dealer. The flow is saved as **Auto Assign Dealer** and activated.

4.7 Milestone 6B - Test Drive Reminder Flow

A Record-Triggered Flow on Vehicle Test Drive runs when a record is created or updated and Status__c equals Scheduled. A scheduled path fires one day before Test_Drive_Date__c, gets the customer information and sends an email to the customer's Email__c with the subject *Reminder: Your Test Drive is Tomorrow!*

4.8 Milestone 7A - Apex Trigger and Handler

The Vehicle Order trigger is configured for before insert, before update, after insert and after update, and calls `VehicleOrderTriggerHandler.handleTrigger`. In the before context the handler retrieves the stock of the related vehicles and blocks the order when Stock_Quantity__c is zero or less, showing: *This vehicle is out of stock. Order cannot be placed.* In the after context, an order with Status__c = Confirmed reduces the stock of its vehicle by one and saves the vehicle record.

```
public class VehicleOrderTriggerHandler {
  public static void handleTrigger(List<Vehicle_Order__c> newOrders,
      Map<Id, Vehicle_Order__c> oldOrders, Boolean isBefore,
      Boolean isAfter, Boolean isInsert, Boolean isUpdate) {
    if (isBefore && (isInsert || isUpdate)) {
      preventOrderIfOutOfStock(newOrders);
    }
    if (isAfter && (isInsert || isUpdate)) {
      updateStockOnOrderPlacement(newOrders);
    }
  }
}
```

4.9 Milestone 7B - Batch Apex

Batch Apex processes pending vehicle orders in manageable batches instead of all at once. For each batch it checks the current vehicle stock, changes eligible Pending orders to Confirmed when stock is available, reduces the vehicle stock by one per confirmed order and saves the updates.

Pending Orders → Batch Query → Stock Check → Confirm Eligible Orders → Reduce Stock → Save Updates

4.10 Milestone 7C - Scheduled Apex

The `VehicleOrderBatchScheduler` class launches `VehicleOrderBatch` with a batch size of 50 at the scheduled interval. The scheduled job can be monitored from Salesforce Setup.

5. Results and Outcomes

Milestone	Main Deliverable	Primary Salesforce Feature
1	Developer environment	Salesforce Developer Account
2	Core data model	Custom Objects
3	Object navigation	Custom Tabs
4	Central application	Lightning App Manager
5	Data capture and relationships	Fields and Lookup Relationships
6A	Automatic dealer assignment	Record-Triggered Flow
6B	Test-drive reminder	Scheduled Path + Send Email Flow
7A	Order validation and stock update	Apex Trigger + Handler
7B	Pending-order processing	Batch Apex
7C	Automatic batch execution	Scheduled Apex

The completed system gives users one Lightning application for all project data and reports, assigns dealers automatically, reminds customers about test drives, prevents out-of-stock orders and keeps vehicle stock consistent with confirmed orders.

6. Conclusion

The project shows how declarative tools (objects, flows) and programmatic tools (Apex, batch and scheduled jobs) can be combined on Salesforce to build a working vehicle order management system. Orders are validated, stock is synchronized and routine customer communication is automated.

7. Future Scope

- Add dashboards and reports for sales, stock levels and service requests.
- Send order-confirmation and service-reminder emails in addition to test-drive reminders.
- Extend dealer assignment with more detailed location rules.
- Add a customer-facing portal for placing orders and booking test drives.